Variable Equation of State Parameter

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Explanation Note on Variable Equation-of-State Parameter in PPC Law of Gravity

In Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity), the gravitational pressure field is defined as:

$$P_g = \omega E_d$$

where:

- P_g = gravitational pressure field,
- E_d = energy density,
- ω = equation-of-state parameter.

The gravitational field force density is defined by:

$$F = -\nabla P_g$$

Substituting the pressure relation:

$$F = -\nabla(\omega E_d)$$

Using the vector product rule for gradients:

$$F = -\nabla(\omega E_d) = -(\omega \nabla E_d + E_d \nabla \omega)$$

This equation shows that the PPC gravitational force density contains two separate physical contributions:

1. **Energy Density Gradient Contribution**

$$-\omega \nabla E_d$$

This term represents the force density generated due to variation in energy density across space.

2. **Equation-of-State Gradient Contribution**

$$-E_d \nabla \omega$$

This additional term appears because the equation-of-state parameter ω is treated as a variable quantity in PPC Law of Gravity.

If ω changes with spacetime coordinates, then:

$$\nabla \omega \neq 0$$

and the second term contributes to the total force density field.

However, when a fixed numerical value of ω is substituted, ω becomes constant locally. Therefore:

$$\nabla \omega = 0$$

and the force density equation simplifies to:

$$F = -\omega \nabla E_d$$

Thus, the generalized PPC formulation allows a dynamical equation-of-state parameter, while the constant-ω approximation naturally removes the coupling-gradient contribution.

This interpretation provides a more flexible gravitational framework in which both energy density variations and equation-of-state variations can influence the gravitational force density field.

$$\omega = \omega(x, t)$$

so spatial or temporal variation of the equation-of-state parameter generates an additional force-density contribution.

> In PPC Law of Gravity, the equation-of-state parameter ω may generally vary across spacetime. Therefore, the gravitational force density contains an additional coupling-gradient contribution proportional to $E_d \nabla \omega$. Under constant-ω approximation, this term vanishes and the force density reduces to $F = -\omega \nabla E_d$.

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